PC assembly from components

Q-1 Install a CPU and a RAM stick onto a motherboard, then apply thermal paste using the pea-sized dot technique before mounting the CPU cooler.

Hint: Take a clear photo of each step as you complete it for your documentation.

ANS:

Steps:

1. Switch off the PC and disconnect the power cable.
2. Place the motherboard on a clean, flat and anti-static surface.
3. Open the CPU socket by lifting the socket lever.
4. Take the CPU by its edges and align the small arrow on the CPU with the arrow on the motherboard socket.
5. Carefully place the CPU into the socket without forcing it.
6. Close the socket cover and lock the CPU using the socket lever. Intel recommends aligning the CPU correctly and handling it by the edges.
7. Locate the RAM slot on the motherboard.
8. Open the RAM slot clips and align the notch on the RAM stick with the notch in the slot.
9. Press the RAM stick down evenly until the locking clips click into place.
10. Check the CPU surface and make sure it is clean.
11. Apply a small pea-sized dot of thermal paste in the center of the CPU.
12. Do not spread the thermal paste manually. The pressure from the CPU cooler will spread it over the CPU surface.
13. Carefully place the CPU cooler on top of the CPU.
14. Secure the cooler using its screws or locking mechanism.
15. Connect the CPU cooler fan cable to the CPU_FAN header on the motherboard.
16. Check that the CPU, RAM and cooler are firmly installed.

Photo Documentation:

1: Motherboard before installing CPU.



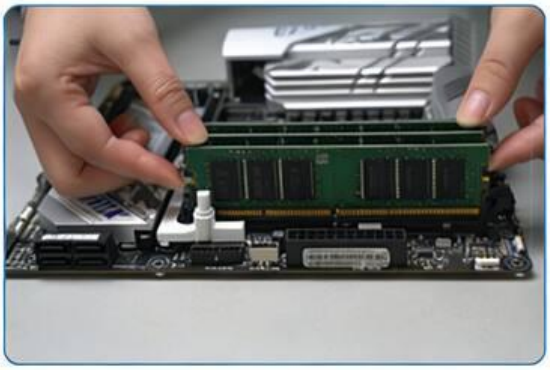
2: CPU correctly placed inside the CPU socket.



1.Open the computer case, check the memory slot position, discharge static electricity from your hands, and clean the dust in the slot



2.Open the latches on both sides of the memory slot

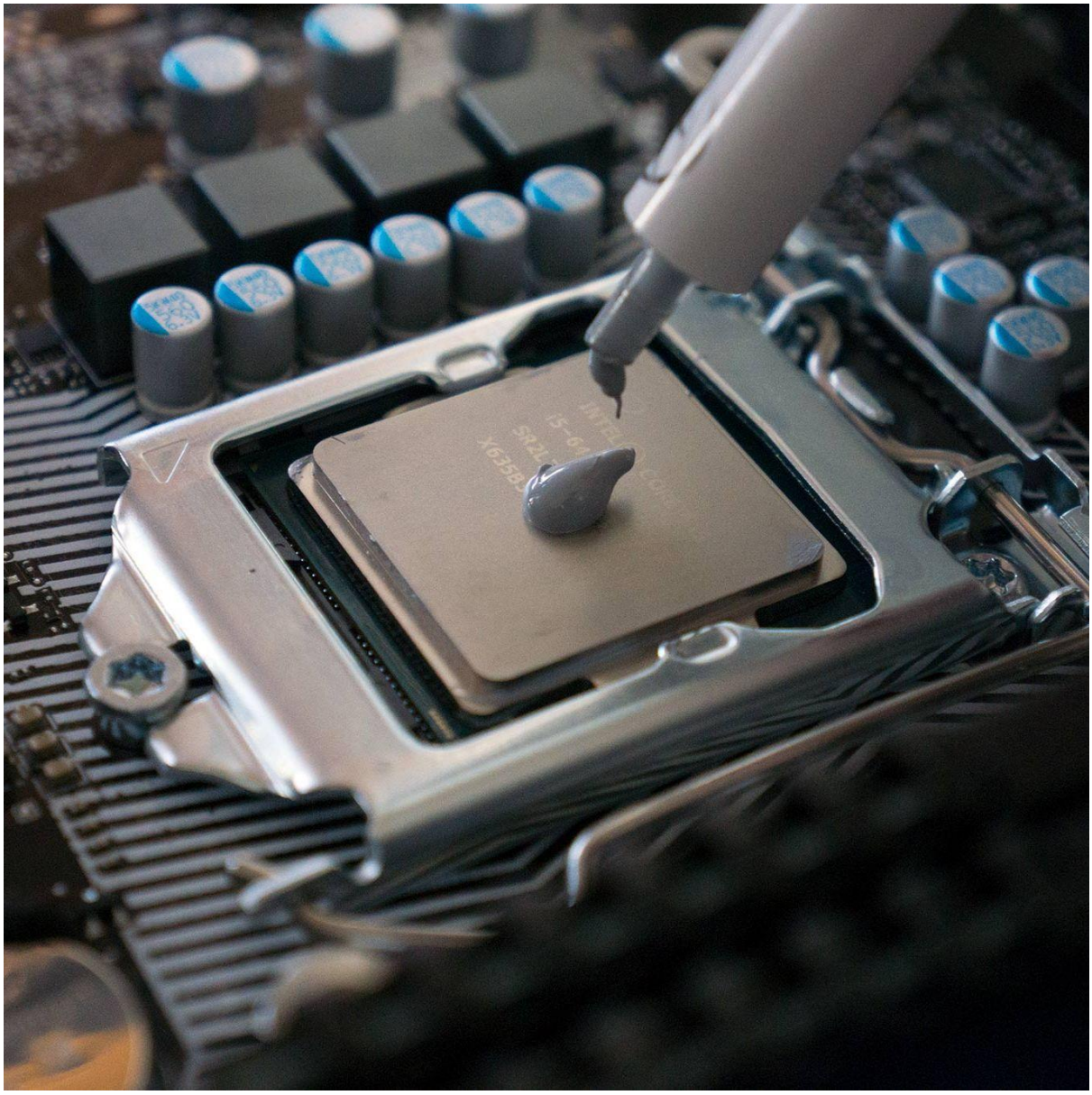


3.Align with the memory key notch, if it doesn't fit, rotate the memory module

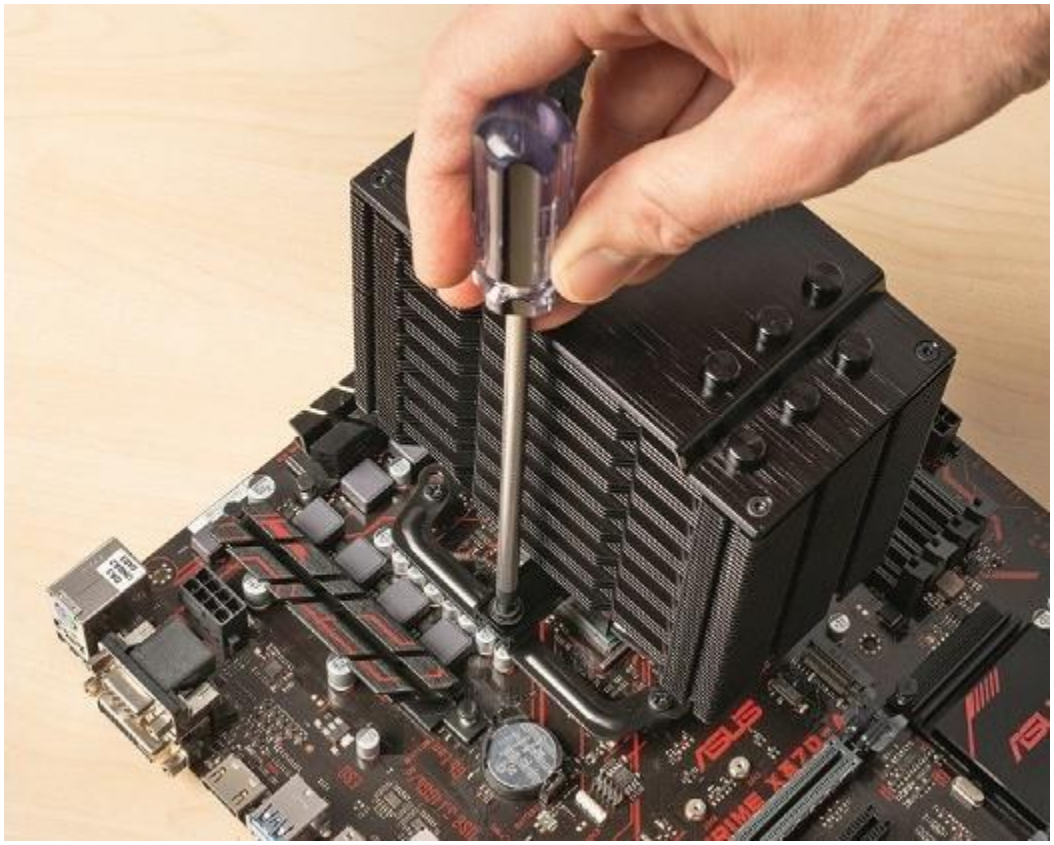


4.After aligning with the key notch, press down on the memory module until a “click” sound is heard, and che c if the golden fingers are exposed. If not, the installation is successful

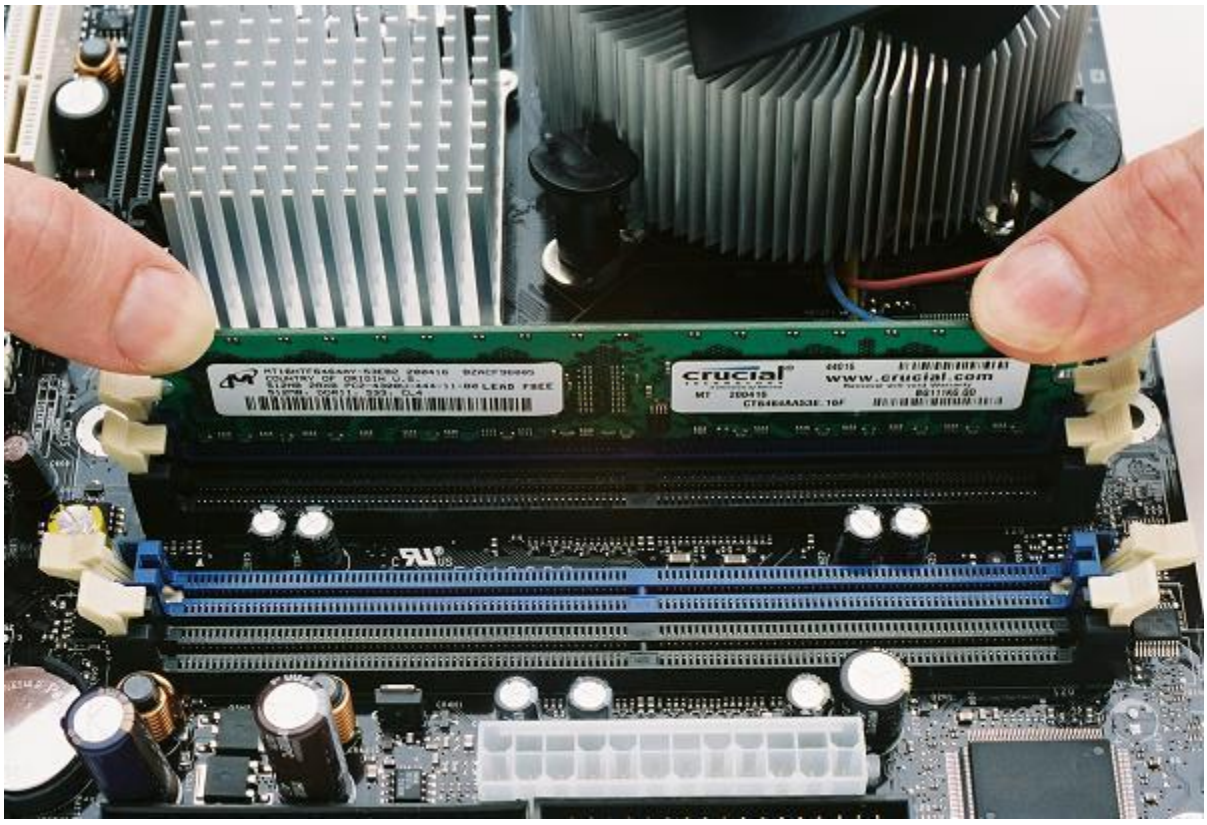
3: CPU socket locked after CPU installation.



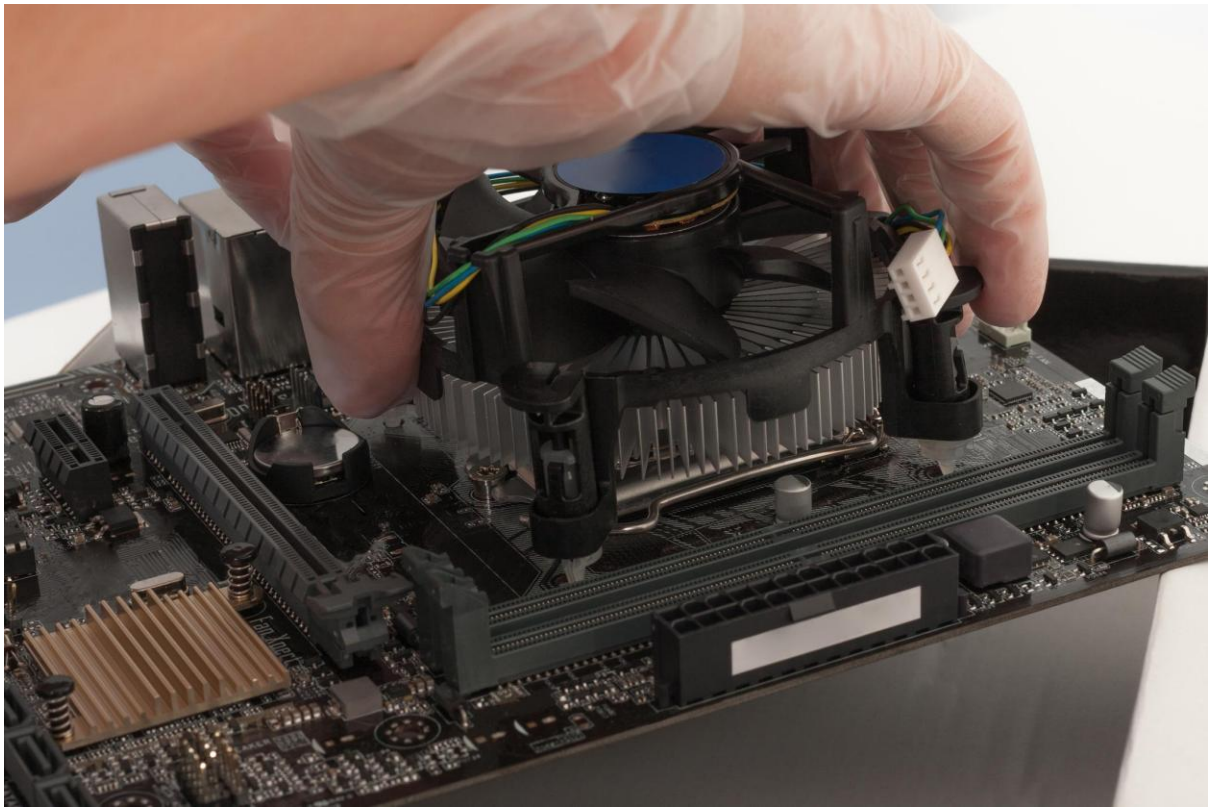
4: RAM stick installed in the RAM slot.



5: Pea-sized thermal paste applied to the center of the CPU.



6: CPU cooler mounted on the CPU.



7: CPU fan cable connected to the CPU_FAN header.

Observation:

The CPU and RAM were installed correctly on the motherboard. A small pea-sized amount of thermal paste was applied to the center of the CPU before installing the CPU cooler. The cooler was then secured properly and its fan cable was connected to the CPU_FAN header.

Q-2 Mount the motherboard inside a PC case and connect both the 24-pin ATX and 8-pin CPU power connectors from the power supply, making sure each connector clicks into place.

ANS:

Steps:

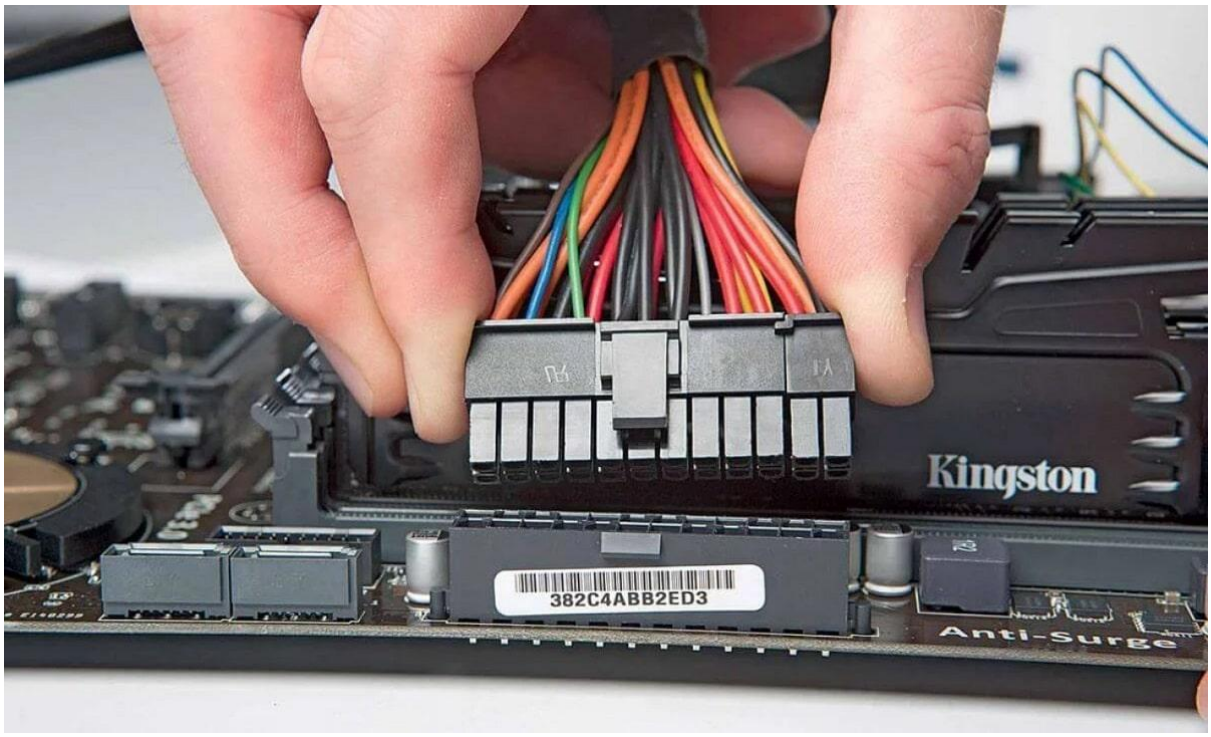
1. Switch off the PC and disconnect the power cable.
2. Place the motherboard inside the PC case carefully.
3. Align the motherboard mounting holes with the standoffs in the case.
4. Secure the motherboard using the required screws.
5. Make sure the motherboard is properly fixed and does not move.
6. Locate the 24-pin ATX power connector on the motherboard.
7. Take the 24-pin power cable from the PSU and align it correctly with the connector.
8. Press the 24-pin connector firmly until the locking clip clicks into place.
9. Locate the 8-pin CPU power connector near the CPU socket.

10. Connect the 8-pin CPU power cable from the PSU.
11. Press it firmly until the locking clip clicks into place.
12. Check both connectors to make sure they are fully inserted and secure.

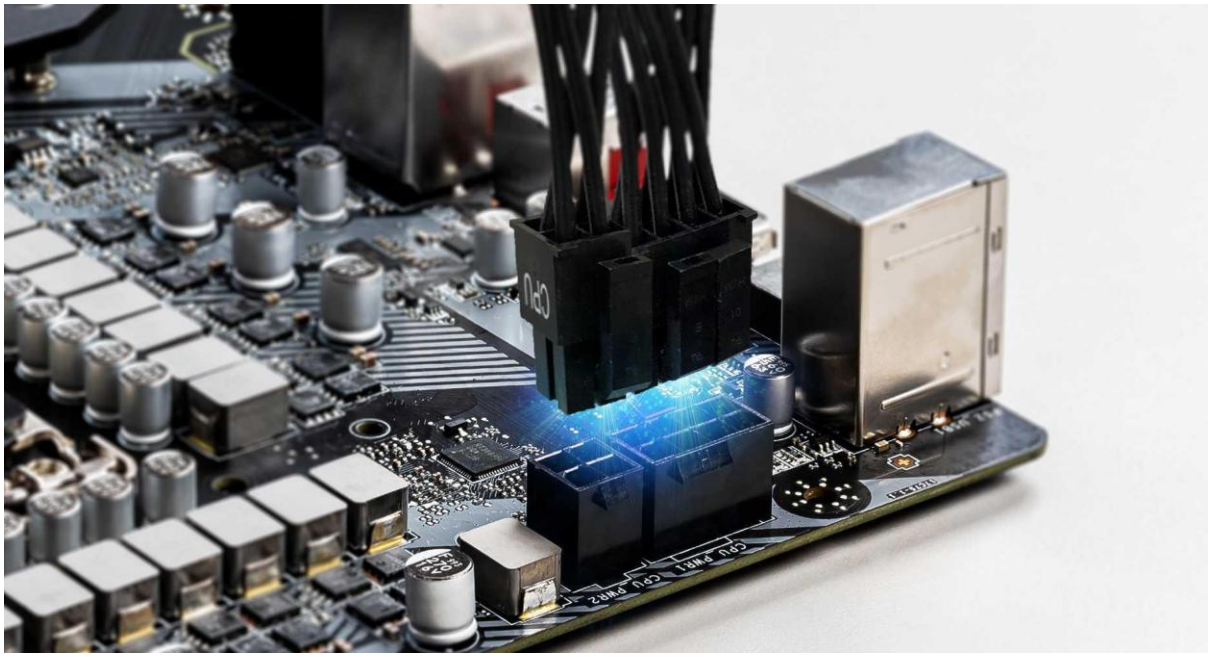
The 24-pin ATX connector provides the main power to the motherboard, while the 8-pin CPU power connector supplies power to the CPU area. The connectors should be correctly aligned and pushed in firmly until fully seated.

Photo Documentation:

1: Motherboard positioned inside the PC case.



2: Motherboard secured with screws.



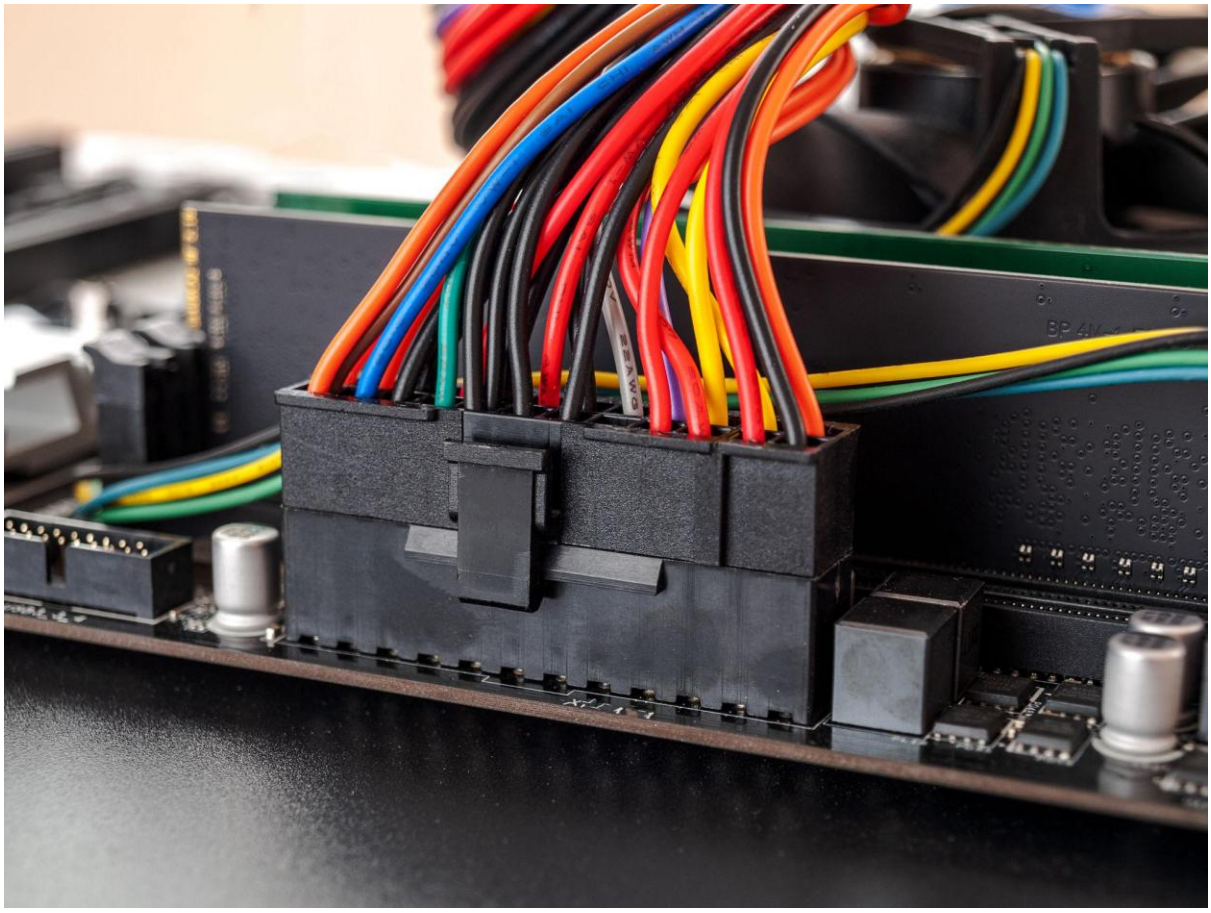
3: 24-pin ATX power connector connected to the motherboard.



4: 8-pin CPU power connector connected near the CPU socket.



5: Final view showing both power connectors properly connected.



Observation:

The motherboard was mounted securely inside the PC case. The 24-pin ATX and 8-pin CPU power connectors were connected from the PSU to the motherboard. Both connectors were pushed firmly until their locking clips clicked into place, ensuring a secure connection.

Q-3 Install a storage drive (SSD or HDD) and a GPU (graphics card) into the appropriate slots, then connect all necessary data and power cables to each device.

ANS:

Steps:

1. Shut down the PC and disconnect the power cable.
2. Install the SSD/HDD in the correct drive bay.
3. Secure the storage drive using the required screws or mounting bracket.
4. Connect the SATA data cable from the storage drive to the SATA port on the motherboard.
5. Connect the SATA power cable from the PSU to the storage drive.
6. Locate the PCIe x16 slot on the motherboard for the graphics card.
7. Remove the required expansion-slot cover from the back of the case.

8. Carefully insert the GPU into the PCIe x16 slot until it is fully seated and the slot clip locks.
9. Secure the GPU to the PC case using the required screw.
10. Connect the PCIe power cable from the PSU to the GPU if the graphics card requires additional power.
11. Check that all storage and GPU cables are firmly connected.
12. Make sure the cables are arranged properly and do not block the cooling fans.

Connection Details:

Storage Drive:

- SATA Data Cable → Storage Drive to Motherboard
- SATA Power Cable → Storage Drive to PSU

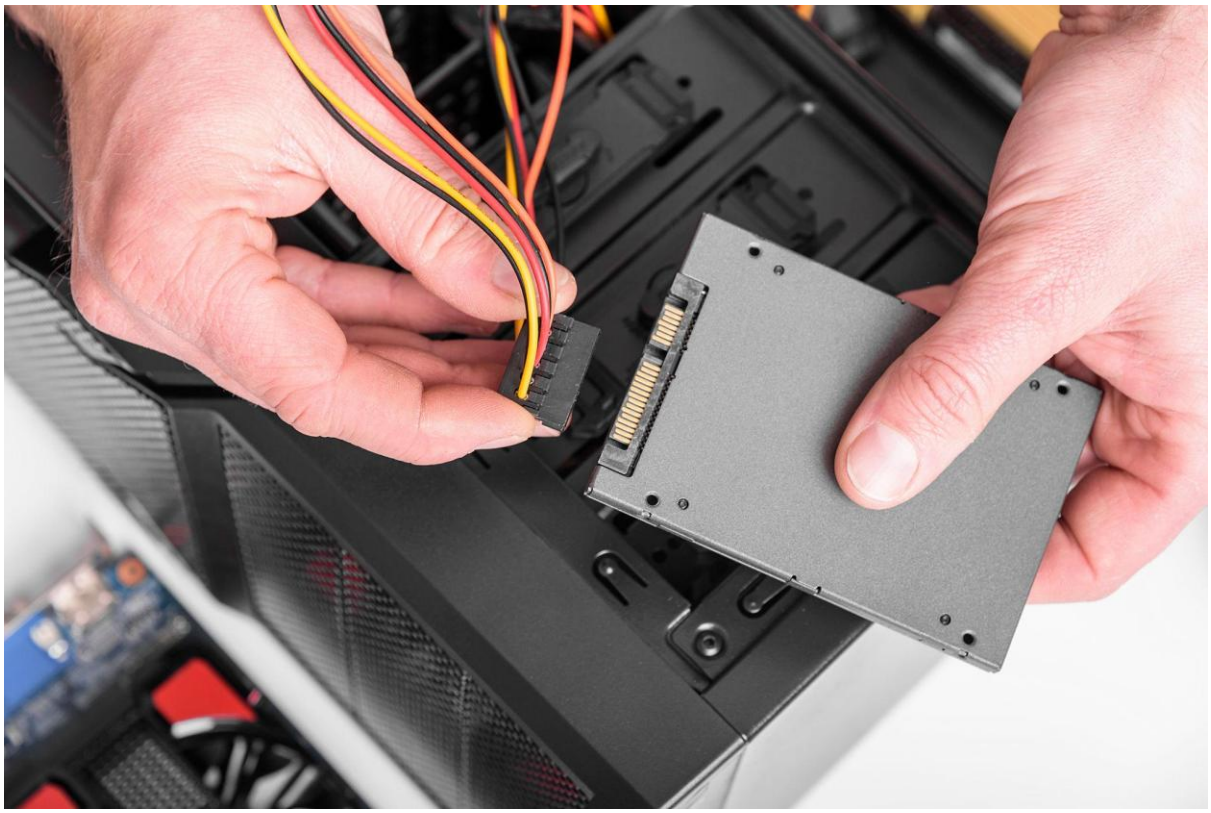
GPU:

- PCIe x16 Slot → GPU to Motherboard
- PCIe Power Cable → PSU to GPU, if required

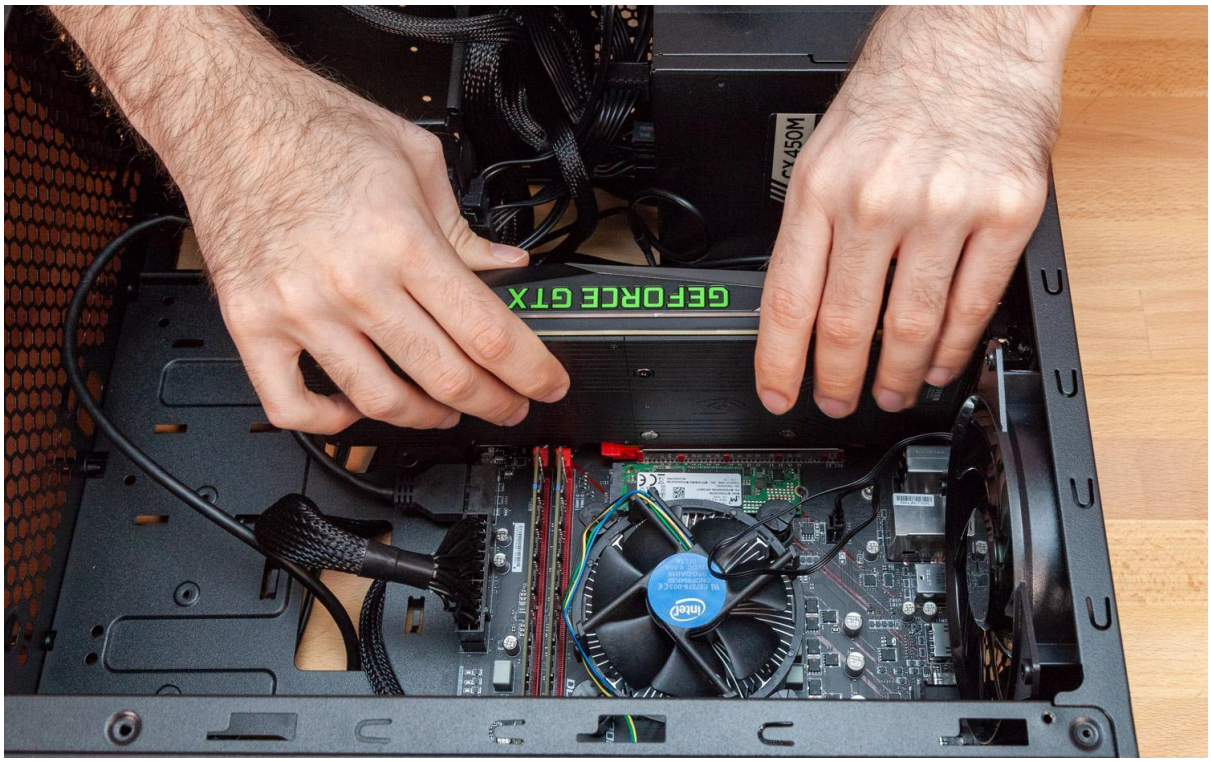
Intel's PC-building guide confirms that a discrete GPU is installed in the PCIe x16 slot and that SATA storage drives connect to the motherboard with a SATA cable and to the PSU for power.

Photo Documentation:

1: SSD/HDD installed inside the drive bay.



2: SATA data cable connected between the storage drive and motherboard.



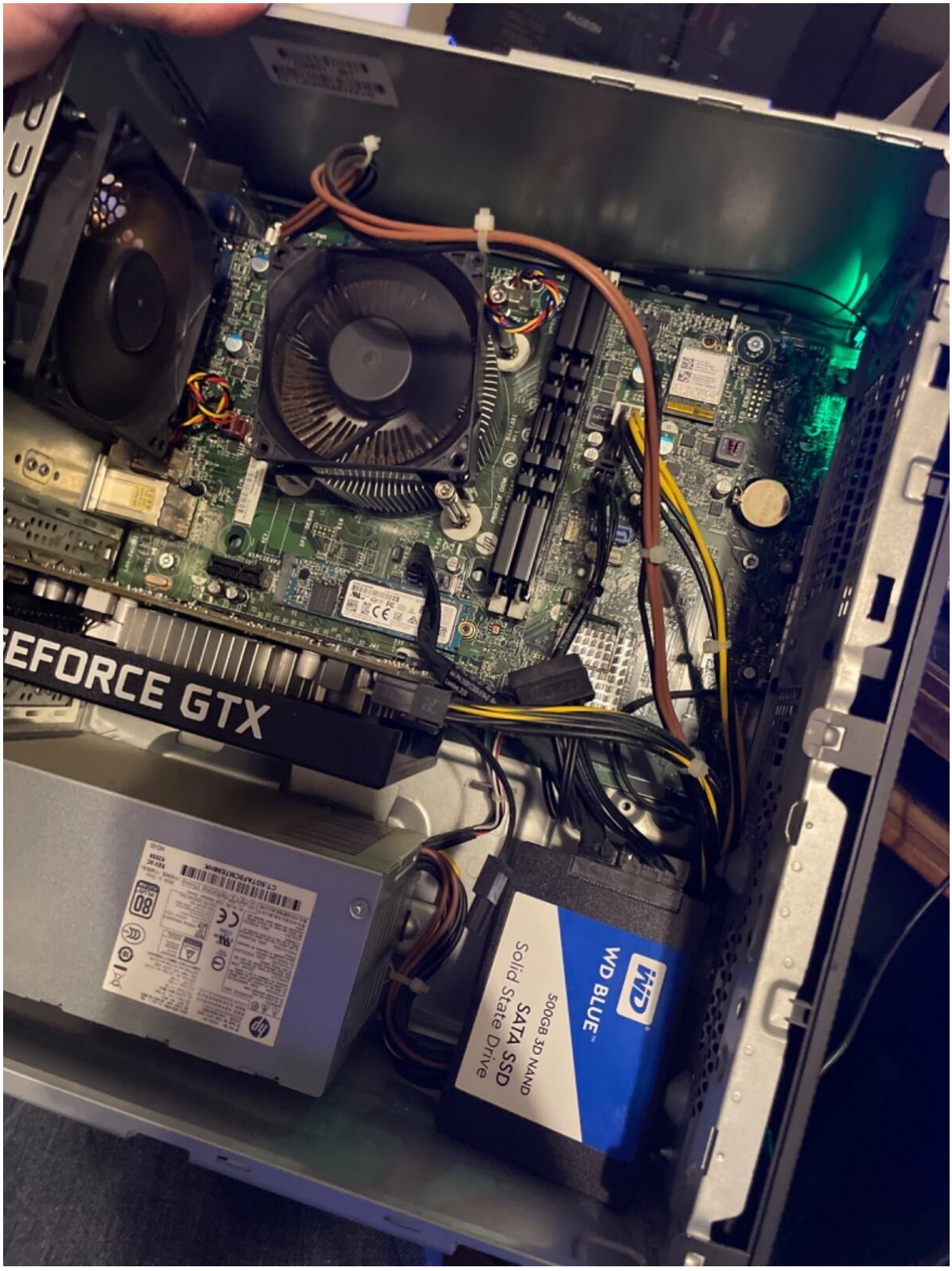
3: SATA power cable connected from PSU to the storage drive.



4: GPU being inserted into the PCIe x16 slot.



5: GPU secured to the case with a screw.



6: PCIe power cable connected to the GPU.

7: Final view showing the installed storage drive, GPU and connected cables.

Observation:

The storage drive was installed securely and connected to the motherboard using the SATA data cable and to the PSU using the SATA power cable. The GPU was installed in the PCIe x16 slot and secured to the case. The required PCIe power cable was also connected to the GPU. All connections were checked to make sure they were secure.

Q-4 Organize and secure all cables inside the case using zip ties or Velcro straps to ensure good airflow and a tidy appearance.

Hint: Take a photo showing your final cable management from both the front and back of the case.

ANS:

Steps:

1. Switch off the PC and disconnect the power cable.
2. Check all cables connected to the motherboard, GPU, storage drive, fans and PSU.
3. Separate the cables according to their connection.
4. Route the cables through the cable-management holes or channels in the case.
5. Keep the main cables such as the 24-pin motherboard cable and GPU power cable neatly together.
6. Use zip ties or Velcro straps to secure the cable bundles.
7. Keep cables away from CPU and case fans so that they do not block airflow.
8. Hide extra cable length behind the motherboard tray or inside the PSU area, if space is available.
9. Check that no cable is loose or touching a fan.
10. Close the case after checking the final cable arrangement.

Cable Management Materials:

- Zip ties
- Velcro straps
- Cable-routing holes/channels
- Cable clips, if available

Observation:

All cables were arranged and secured neatly using zip ties/Velcro straps. Extra cables were routed behind the motherboard tray where possible. The cables were kept away from the fans and airflow paths, making the inside of the case cleaner and helping maintain good airflow. Cable-routing channels and Velcro/zip-tie points are commonly provided in PC cases for this purpose.

Photo Documentation:

1: Front View – Final Cable Management



2: Back View – Final Cable Management



3: Close-up of Zip Ties/Velcro



Q-5 Perform a final verification: check for any loose screws, confirm all power/data connections are firmly attached, and document your completed build with a series of labeled photos showing each major component installed.

ANS:

Steps:

1. Switch off the PC and disconnect the power cable before doing the final internal check.
2. Check the motherboard and make sure there are no loose screws or unwanted metal parts inside the case.
3. Check that the CPU cooler is properly fixed and the CPU_FAN cable is connected.
4. Check that the RAM stick is fully inserted and the locking clips are closed.
5. Check the 24-pin motherboard power connector and 8-pin CPU power connector.
6. Check the GPU is fully inserted into the PCIe slot and secured with a screw.
7. Check the GPU power cable, if required.
8. Check the storage drive and make sure its data and power cables are firmly connected.
9. Check all case-fan cables and front-panel cables.
10. Check that no cable is touching or blocking any fan.
11. Check that all components are securely mounted.

12. After completing the inspection, connect the power cable and switch on the PC to confirm that it starts normally.

Final Verification:

All major components were checked after completing the PC assembly. No loose screws or loose cables were found. The motherboard, CPU, RAM, storage drive, GPU, PSU and cooling system were securely installed. All required power and data connections were checked before powering on the PC. Checking the 24-pin, CPU power, GPU power and storage connections is an important part of final PC troubleshooting.

Photo Documentation:

1: Motherboard Installed

2: CPU and CPU Cooler



3: RAM Installed

4: Storage Drive



5: GPU Installed

6: PSU Connections



7: Cable Management



8: Final Completed Build



Final Observation:

The PC build was inspected carefully after completing the assembly. All major components were securely installed, power and data cables were firmly connected, and the cables were arranged neatly. No loose screws or unwanted objects were found inside the case. The completed build was documented with labeled photos of each major component.